SEQUENCE AND COLLABORATION DIAGRAM

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INTERACTION DIAGRAM

- models describe how groups of *objects* collaborate in some behavior
- NOTE THAT: not class
- typically, captures the behavior of a single use case
- Two kinds
  - sequence diagrams
  - collaboration diagrams

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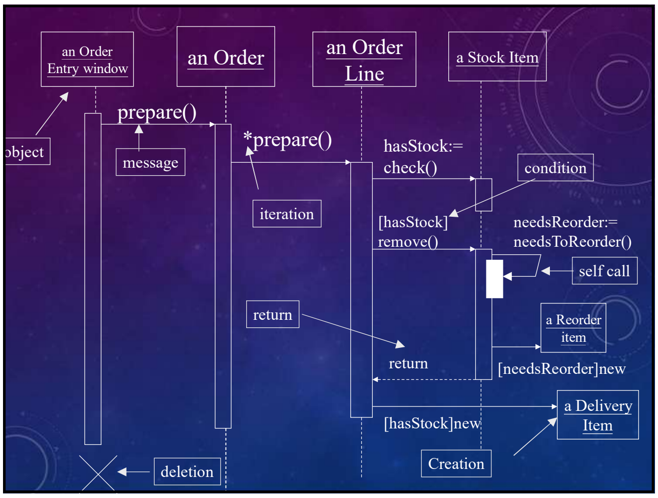

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WHEN TO USE SEQUENCE DIAGRAM

- One of the hardest things to understand in an object-oriented program is the overall flow of control.
- A good design has lots of small methods in different classes and at times it can be tricky to figure out the overall sequence of behavior
- Sequence diagram helps you see the sequence
- valuable for concurrent process

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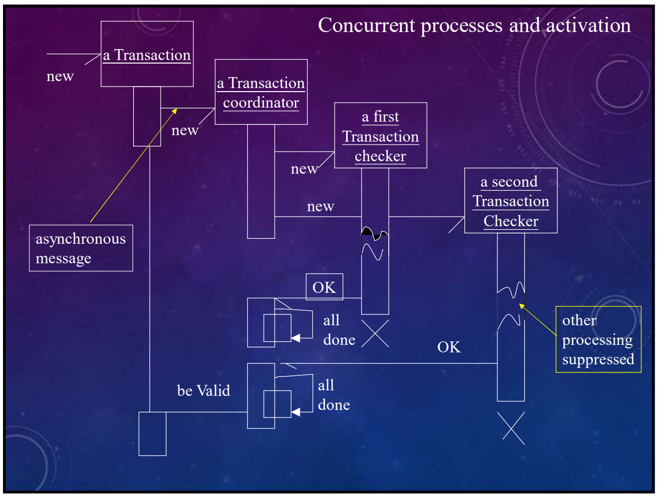

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SEQUENCE DIAGRAM

- 請看一下下面兩張sequence diagram

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Figure 4.1. A sequence diagram for centralized control

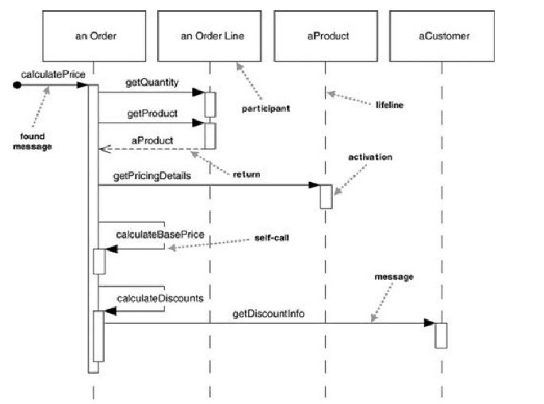

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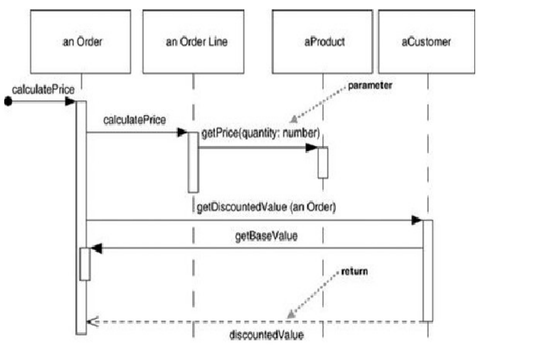
Figure 4.2. A sequence diagram for distributed control



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COMMENTS

- Sequence diagram 並不擅長展示有 loop 等的演算法
- 適合展示某個複雜的互動情境

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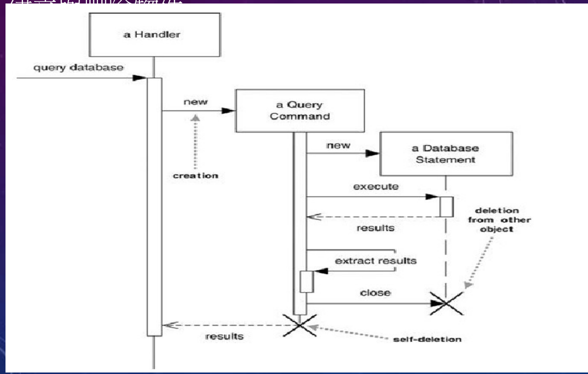
建立與刪除物件



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Loops, Conditionals, and the Like

A common issue with sequence diagrams is how to show looping and conditional behavior. The first thing to point out is that this isn't what sequence diagrams are good at. If you want to show control structures like this, you are better off with an activity diagram or indeed with code itself. Treat sequence diagrams as a visualization of how objects interact rather than as a way of modeling control logic.

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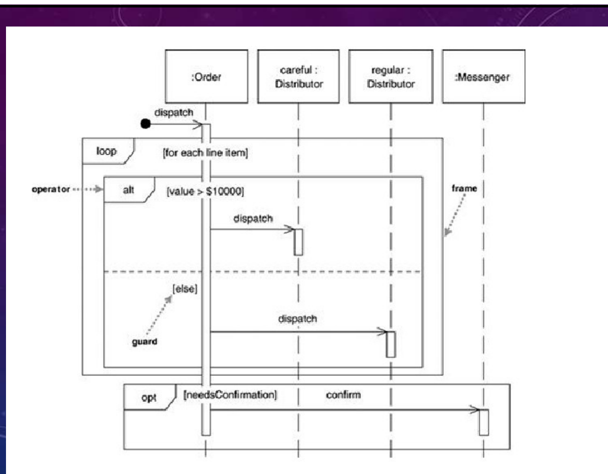

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procedure dispatch
foreach (lineitem)
if (product.value > $10K)
careful.dispatch
else
regular.dispatch
end if
end for
if (needsConfirmation) messenger.confirm
end procedure

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LOOP AND ALTERNATIVE FRAME

- Loop 以及 alternative 框架是 UML 2.0 之後才有的
- 有些人不喜歡 loop 以及 alternative 框架，而自行採用其他的方法如下圖

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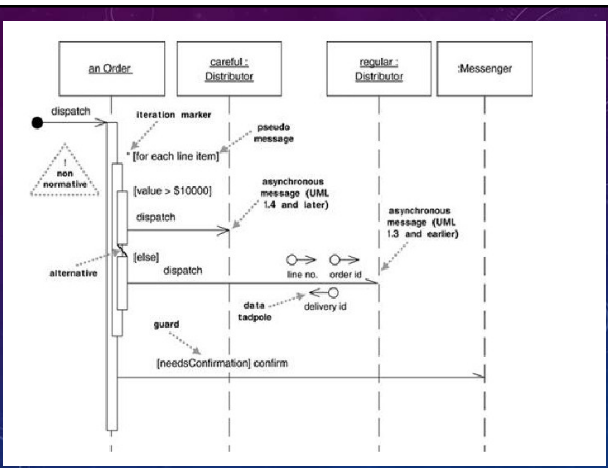

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FRAME OPTION

- alt Alternative multiple fragments; only the one whose condition is true will execute (Figure 4.4).
- opt Optional; the fragment executes only if the supplied condition is true. Equivalent to an alt with only one trace (Figure 4.4).
- par Parallel; each fragment is run in parallel.
- loop Loop; the fragment may execute multiple times, and the guard indicates the basis of iteration (Figure 4.4).
- region Critical region; the fragment can have only one thread executing it at once.
- neg Negative; the fragment shows an invalid interaction.
- ref Reference; refers to an interaction defined on another diagram. The frame is drawn to cover the lifelines involved in the interaction. You can define parameters and a return value.
- sd Sequence diagram; used to surround an entire sequence diagram, if you wish.

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COLLABORATION DIAGRAM

- The sequence is indicated by numbering the message
- the spatial layout allows you to show other things more easily
- show how object link together
- you can overlay packages or other information more easily

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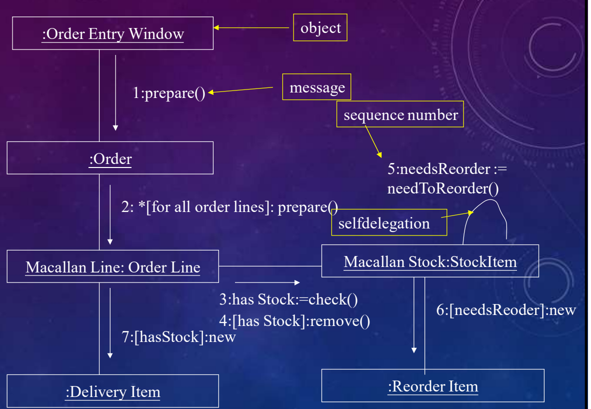

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SEQUENCE OR COLLABORATION?

- different people have different preference for different situation
- interaction diagrams can awkward to use when exploring alternatives

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HOW TO FIND METHODS (BEHAVIORS, OPERATIONS) OF OBJECT

- 好，你應該知道要怎麼樣以sequence diagram描述物件之間的互動
- 重點是，你的物件的 method 根本還沒有個譜？
- 你怎麼找他們？

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9.2. THE KRB SEVEN-STEP METHOD.

STEP ⑦ OPERATIONS (I.E., BEHAVIOR)

Six techniques for Finding Operations

1. By Inspection
2. Basic *CRUD*
3. ~~Use Cases~~
4. Statechart Diagram
5. CRC Cards
6. *CRUD* Revisited

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WHAT IS A USE CASE？

- 使用案例（系統使用情境 scenario）
- Use cases 是你的系統所要完成的功能
- 這些功能當然要由你所設計的 objects 互動來完成
- 請看上次的規格書

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Figure 9.1. Example use case text

Buy a Product

Goal Level: Sea Level

Main Success Scenario:

1. Customer <u>browses catalog and selects items to buy</u>
2. Customer goes to check out
3. Customer fills in shipping information (address; next-day or 3-day delivery)
4. System presents full pricing information, including shipping
5. Customer fills in credit card information
6. System authorizes purchase
7. System confirms sale immediately
8. System sends confirming e-mail to customer

Extensions:

3a: Customer is regular customer
 .1: System displays current shipping, pricing, and billing information
 .2: Customer may accept or override these defaults, returns to MSS at step 6

6a: System fails to authorize credit purchase
 .1: Customer may reenter credit card information or may cancel

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WHAT IS A SCENARIO (情境）

A **scenario** is a sequence of steps describing an interaction between a userand a system.

So if we have a Web-based on-line store, we might have a Buy a Product scenario that would say this:

The customer browses the catalog and adds desired items to the shopping basket. When the customer wishes to pay, the customer describes the shipping and credit card information and confirms the sale. The system checks the authorization on the credit card and confirms the sale both immediately and with a follow-up e-mail.

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WHAT IS A USE CASE?

- 一個使用案例是一組（系統使用，或系統操作）情境來達成某個目標
- 一個趨勢是規格書的寫法漸漸由 use cases 所取代，尤其是軟體代工產業

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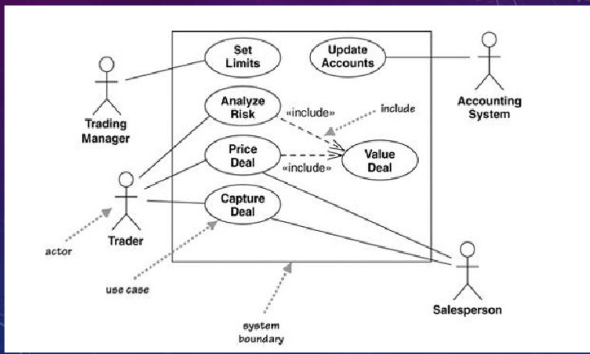
USE CASE DIAGRAM



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USE CASE干尋找物件METHOD啥事

- 一旦有了 use case (可以有正式文件，或者是 建模者自行寫下一些scenario
- Scenarios (use cases) 是系統行為的起頭
- 每一個use case 最終一定要由系統中的物件來完成
- OK， 接著？

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